Techno-meter

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Abstract

Egypt has faced Many challenges that hinder its progress throughout the last century. Various aspects of life are affected by these challenges and circumstances; Ascending rate of the produced garbage and wastes represents part of these challenges impact on environment and human beings. E-wastes may be the most hazardous because of including various metals and semi-conductors which also are valuable to be recycled, so our chosen solution was reusing them in scientific field to be more beneficial if they are exploited properly. As a result these wastes have been used in a better way in the design of our solution which is a speedometer that is built by E-wastes recycled from discarded electric device around us , the design requirements of this solution includes being used in more than one lab or being multi-functional , it mainly depends on a recycled microcontroller from old smart camera and an old toy car with some sensors that are connected to the microcontroller .Our conducted tests for these requirements were performed according to our test plan, some results from these tests weren't satisfying so some adjustments were made then our required results were met , we analyzed these results to check our solution outcome , ultimately it has been shown that our project can perform its function as a speedometer which measure car speed then draw graphs on a designed website by us.

Introduction

Egypt is currently facing 11 great obstacles that hinder its flourishing, among them are improving the scientific and technological environment (S.T.E.) for all, addressing and reducing pollution that fouls our air, water, and soil, adapting to and mitigating the effects of climate change, and recycling waste for economic and environmental purposes. These Egyptian challenges are a significant issue due to their impact on the chances of achieving sustainable development. Egypt ranks first worldwide for the number of deaths attributable to PM2.5 air pollution **(Ghanem et al, 2023).**

Moreover, research shows that after investigating the water quality of 24 sites between Aswan and Cairo along the Nile to determine water quality (WQI), heavy metal pollution (HPI), and contamination (Cd), water quality deteriorated significantly. These heavy metals are directly associated with electronic waste; for example, the excessive amounts of copper, iron, tin, and silicon in the Nile River ,This is directly linked to the ineffective E-waste recycling strategy Egypt currently follows.

The central problem behind these issues, which the project aims to resolve, is the current lack of recycling and the disposal of potentially valuable devices that could have improved the S.T.E., which were labeled as "waste". Some solutions and techniques were generated before our project; among them are PCB (printed circuit board) recycling and the ultrasonic parking sensor.

Firstly, the PCB recycling process generally involves three critical stages: pretreatment, physical recycling, and chemical recycling. The only downsides are the method's complexity and the need for efficient, scalable recycling technologies to mitigate the growing e-waste problem**(Q Tan et.al, 2020).** Secondly, the ultrasonic parking sensor relies on the reflection of sonic waves to measure the distance, and with some further adjustments, it can measure the speed and acceleration if paired with a microprocessor. Some of its merits are its accuracy, as it measures the exact value of the distance, and being easy to connect; however, its only demerit is having low precision, as it is affected by external factors like humidity and air density **(Jeon et al,2019)**.

The project's requirements were to be primarily constructed from old or discarded devices or components, execute either two functions in a single lab or a single function in two labs, serve the biology, physics, chemistry, geology, or the fabrication Lab, and the error percentage of the prototype has to be **less than ±5%,** be free of any hazardous or radioactive materials.

The solution chosen was to construct a contraption that is made with recycled E-waste components, serves the education process, and contributes to the improvement of the scientific and technological environment. The chosen solution adheres to the design requirements and constraints. So, our project aims to serve the physics laboratory using a **discarded toy car**, an **ESP32 CAM** as a microcontroller, and an **ultrasonic sensor** to measure the distance between the car and a fixed barrier, the speed of the car, its acceleration, and to display these data on an **IoT**-integrated dynamic website.

Materials & Methods

Materials :

Item	Usage	Picture	Item	Usage	Picture
Esp32cam	From an old camera and it is the mind of the project		Old toy car	We put inside it all the components	
Ultrasonic sensor	We used to measure distance		Old track from wood	We used this track to put the car on it	
Jumber wires	To connect components		Micro usb cable	To transfer code to esp32cam	
Iron naits	For the wood track		Wax gun	To stick the components	

Methods

The construction of the prototype (speedometer) requires achieving design requirements and configuring the electric circuit first, we make a schematic diagram of the project to simulate the prototype and its electric connections.



Steps of installation:

1. A deep search was conducted to find old electric devices that include a microcontroller.
2. After locating an old smart camera that includes microcontroller called esp32 cam, the microcontroller was extracted from it to be the "mind" of our contraption.
3. A calibrated ultrasonic sensor extracted To detect the distance covered by the body.
1. Old jumper wires were used to connect these components to each other.
2. The microcontroller was coded to take the reads from the sensor to return the results of the sensor detection.
3. Programmed a website with (HTML, java, CSS) to show the results in various ways such as tables and graphs.
4. Stuck all these components on the body of an old toy car to be the detected moving body
5. Design a small track with old wood Remnants its length about 122 ± 2cm it will be the road of the moved body (toy car) in our prototype.



Figure 1

Test plan:

1. Every process in the test plan has been done through safety regulations and right precautions such as putting on masks and gloves, wearing lab coats, avoid getting hurt while using any dangerous equipment.

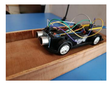


Figure 2

2. Calibrating ultrasonic sensor according to the location conditions.
3. Attaching the whole system to the moving body (old toy car)
4. Putting the car on the track then turning it on
5. Checking the website before starting the test
6. Moving the car with the remote-control forward in a straight way on the track
7. Taking the results from the website to be analyzed



Figure 3

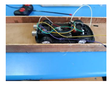


Figure 4

Results

- **Negative results:**

After executing our tests according to the test plan, initially some negative results came out, as there were some issues with the used hall effect sensor, as it either skipped a whole revolution of the magnet attached to the spinning wheel and it was not compatible with our microcontroller, which affected the results' accuracy.

- **Positive results:**

So, after further failed trials, using other sensor, which was the ultrasonic sensor, was the best option to generate accurate results. Fortunately, the ultrasonic sensor returned accurate results with utter precision, as the error percentage was **±2%**, as by measuring the length of the track manually **5 trials** we found the average is **1± 0.02 m** and the system returned data exhibited that the average distance travelled of 5 trials **was 0.98± 0.02 m** As shown in **Table (1)**

Table 1: Distance readings

Distance	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average
Manual measuring value	1.02 m	1 m	1.01 m	0.98 m	0.99 m	1m
Measured value by sensor	1 m	0.98 m	0.99 m	0.96 m	0.97	0.98 m

Then the other quantities such as speed which calculated according to **5 trials** readings of time and distance and acceleration are calculated from the conducted graphs through website. As shown in **Table (2)**

Table 2: Test Plan Readings

Quantity	Trial 1	Trial 2	Trail 3	Trial 4	Trial 5
Distance	1 m	0.98 m	0.99 m	0.96 m	0.97 m
Time	1.72 s	1.67 s	1.75 s	1.6 s	1.5 s
Avg. Speed	0.581 m/s	0.587 m/s	0.566 m/s	0.600 m/s	0.647 m/s
Avg . Acceleration	1.5	1.48	1.56	1.6	1.58

Analysis

On spotting the light on Egypt's grand challenges, especially recycling and reused electrical devices, these two are key drivers of environmental problems, pollution, public health risks, and a variety of other challenges that confront Egypt. In the process of recycling, our speedometer is effective as it helps us to reuse old components to construct a new device and allow for better scientific and technological improvement. Speedometers support physics labs such as measuring speed and acceleration and distance and time. Our speedometer reduces electric waste and enhances the efficiency of technological environment ; it may be possible to reduce climate change. Most regions and countries on the earth have problems with climate change , recycling and pollution that come from electrical waste. Because of this most people have been affected by these. Every prototype has problems, but our prototype has demonstrated a combination of exploiting strong points and avoiding week points. We achieved results that go beyond most of our design requirements. Our prototype has strength points like low cost and its parts easily to repair and enhance the educational system operations , as it has contributed to deep understanding of some physics concepts such as speed and acceleration ,also we made a website for our project, and it has all results to give all information.

Old remote-controlled car:

As shown in Figure 1,2,3,and 4 an old toy RC car was dismantled to be the test subject of the project, of which acceleration, distance covered, and speed will be tested.



Figure 5

The ESP32 CAM microcontroller includes Wi-Fi and Bluetooth functionality, Bluetooth Low Energy (BLE) communication, independent timers, analog-to-digital and digital-to-analog converters (ADCs and DACs), **as shown in Figure 5**. The ESP32 microcontroller includes two 240MHz cores, each with a Tensilica Xtensa 32-bit LX6 microprocessor (**Cameron, 2023).** The ultrasonic sensor is a non-contact sensor that includes 4 main pins, VCC (5V) , Trigger, Echo, and GND (Ground) respectively **as shown in Figure (6),** Measures distance with utter accuracy. However, the precision of its measurements is often low, as it is affected by external factors like humidity and air density. The TOF (time of flight) is the time taken by an ultrasonic wave to hit an obstacle and return. Ultrasonic-distance measurement with the TOF method is based on the following physical parameters: speed of sound in the air, environmental factors, and time of detection of the reflecting **wave as illustrated in Equation 1.**

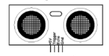
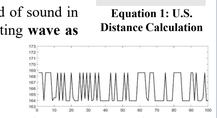


Figure 6

$$D = \frac{V_S \cdot T}{2}$$

Equation 1: U.S. Distance Calculation

As mentioned above, there is a typical issue with using the ultrasonic sensor, which displays a stable graph without sudden dips and peaks, as exhibited in graph 1, which is meant to represent a static object, which reduces the overall precision.



Graph 1: Imprecise U.S. Readings

Equation 2 explains the change in sound speed due to temperature. C is the temperature in Celsius. Humidity impact is usually negligible in practice, but temperature must be taken into consideration to measure the distance changes up to the millimeter level. If the temperature changes by about 5 degrees Celsius, the distance by about 3 mm in 1000 microseconds (the speed of sound is 342.3 m/s at 20 degrees Celsius, and the sound distance taken is about 343.2 mm in 1000 microseconds) **(Aliew, F. 2022).**

$$V_s = 331.2 + (C \times 0.6)$$

Equation 2: Speed Formula

A second test plan was conducted; however, the given result of 0.83m travelled distance was 0.81 m, as shown in graph 2, which is a 2% error percentage. This happened because the external factors, like humidity, temperature, and wind speed, were taken into consideration as windows were closed and fans were turned off to reduce the noise that could disturb the accuracy.



Graph speed

The system also ,After conducting the first test plan with the use of the hall effect sensor, which proved its ineffectiveness, as a measuring tape was laid on the table to measure the distance covered by the toy car, which was one meter, the sensor returned to a value of 0.7m. The reasons behind this 30% are that the Hall effect sensor

$$C_i = 2\pi r,$$

Equation 3

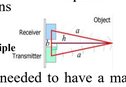
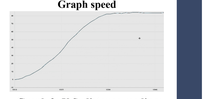


Figure 7: U.S working principle



Graph 2: U.S. distance readings

was incompatible with the project overall, as it needed to have a magnet attached to the spinning wheel of the RC car to calculate the number of turns by counting the magnet pulses and, with it, determine the distance covered, however, a small magnet was bigger that it should **be as shown in Equation 3** which is used to calculate the distance covered per revolution, which is the circumference of the wheel, the magnet's size made the sensor falter, as it sometimes skipped a pulse or did not complete a whole revolution. Two smaller magnets were attached in a desperate attempt to solve this issue, but with no success. Changing the sensor was the only option to reduce the error percentage.

The ultrasonic sensor was the best option as it has accuracy by sensing the reflection of sonic waves that are cast from the transmitter as they bounce off an object and return to the receiver using basic geometry, **as shown in Figure 7 and Equation 3 (Zhmud et. al,2018).**

measured the velocity and **acceleration 2% error** percentage, as both of them are dependent on the measured distance and **time as shown in equation 4 and 5,** and shows the accuracy of the ultrasonic sensor.

$$a = \frac{\Delta v}{\Delta t}$$

Equation 5: Acceleration Formula

$$v = \frac{d}{t}$$

Equation 4: Velocity Formula

Table 3: Learning Transfer

Subject	Learning Transfer
PH.2.03	Clarified the laws of electrodynamics and helped us determine the suitable voltage for each component using Ohm's law (V=IR).
PH.2.06	Facilitated the usage of the Hall effect sensor, as it discussed magnetic fields and helped in determining the suitable distance between the sensor and the magnet.
English	Help in conducting this research as a whole, in writing the poster and portfolio academically.
MA.2.01	Facilitated the graphing of the equations used and ensured the correctness of the results.
CS.2.01	Aided in determining the topological shape and connecting the website to the contraption via local cloud.

Conclusion

This research tackles not only the rising concern regarding the disposal of electronic waste but also the incompleteness of accessible and reliable solutions in terms of modeling motion ideas in a physics lab in schools. The use of recycled parts from electronic trash coupled with the ESP32 CAM microcontroller proves that engineering through sustainability can indeed create educational technologies that are both effective and affordable. The analysis of the ultrasonic sensing function regarding the time-of-travelled distance calculation provides the essential theoretical background in forecasting accuracy with remarkable consistency in terms of speed and acceleration. The analytical results are well supported by the experimental testing. The ultrasonic sensor yielded consistent and comparable results in the laboratory setting, and the calculated values of distance, speed , and acceleration closely matched the theoretically expected values of motion in the track. This confirms that this system provides effective real-time data that can be utilized in physics lab works. This current prototype compares favorably with the former solution that relied upon commercial motion sensors that are too expensive and unrepairable when compared with this environmentally responsible and affordable system that functions correctly and remains straightforward in terms of maintenance. The results obtained confirm the effectiveness of the use of recycled electronic components in satisfying the requirements necessary for powering a complete and accurate motion-measurement system, ensuring physics lab access enhancement and reduction of e-waste. Based upon the confirmation from both analytical and experimental results, the next level involves utilizing this information in designing recommendations.

Recommendation

There is no perfect prototype, and our prototype is not exception. We recommend some features and attributes that can make major effects in the project that could help in improving the application and efficiency of our project. To turn this project to a final product several suggestions should be considered as follows:

I. Improve Measurement Accuracy: using better, more accurate sensors along with the processing unit is critical for a larger scale building, considering its scarcity regarding small scale prototypes.
II. Reinforce Mechanical Components: Strengthen the car chassis and overall car body and the track, ensure stable performance and reduce errors considering high-speed testing
III. Adopt adaptive design: use flexible sensor and hardware modules for replacements and customizations for educational lab experiments.
IV. Increase Sustainability Impact: adopt a well-studied technique of recycling waste materials (using a well-studied plan of which materials will be recycled and how to ensure having the best impact on earth)
V. Ensure Reliability and Safety: to make a final product out of our prototype, it must meet several safety requirements and standards to withstand any crisis and save students from any fatal error

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